

Protocol A

1B@32k	-0.050 MAE 0.254 median	not evaluated	-1.635 MAE 2.272 strength	-0.072 MAE 0.103 bit median	+0.388 MAE 1.050 bit median
1B@96k*	+0.071 MAE 0.193 median	not evaluated	-1.029 MAE 1.302 strength	+0.037 MAE 0.074 bit median	+0.223 MAE 0.397 bit median
1B@112k	-0.018 MAE 0.191 median	not evaluated	-1.087 MAE 1.398 strength	-0.035 MAE 0.071 bit median	+0.125 MAE 0.676 bit mean
6.9B@32k	-0.316 MAE 0.476 zero	not evaluated	-1.360 MAE 1.635 median	-0.220 MAE 0.253 zero	-2.138 MAE 3.972 bit median
6.9B@112k	-0.344 MAE 0.436 median	not evaluated	-1.931 MAE 2.121 strength	-0.229 MAE 0.271 bit median	-3.443 MAE 3.873 bit mean
410M@48k	not evaluated	-0.092 MAE 0.386 median	not evaluated	not evaluated	not evaluated
1.4B@112k	not evaluated	-0.218 MAE 0.402 median	not evaluated	not evaluated	not evaluated
6.9B@80k	not evaluated	-0.136 MAE 0.370 median	not evaluated	not evaluated	not evaluated
Prune interp d=.9-.6 Prune interp d=.85/.675/.575 Prune extrap d=.55 Quantization bits ≥4 Quantization int3					

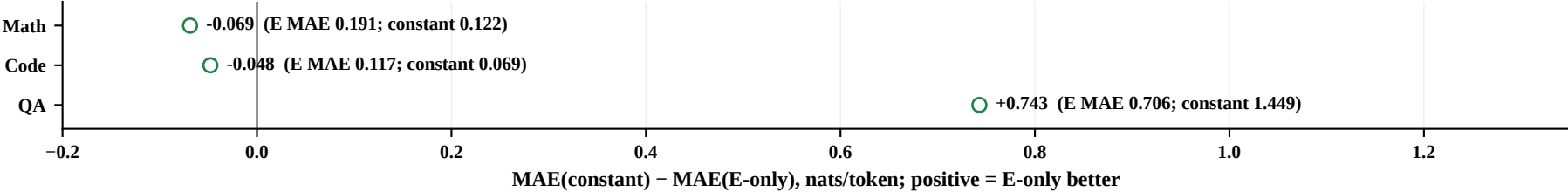
Protocol B

-0.025 MAE 0.249 median	-0.241 MAE 1.772 strength	-0.002 MAE 0.028 bit median	+0.021 MAE 0.606 bit mean
not evaluated	not evaluated	not evaluated	not evaluated
-0.028 MAE 0.223 median	+0.377 MAE 0.957 strength	-0.000 MAE 0.038 bit mean	+1.986 MAE 1.120 bit mean
-0.024 MAE 0.160 median	-0.557 MAE 0.705 median	-0.008 MAE 0.032 bit median	-0.315 MAE 0.489 bit mean
-0.035 MAE 0.131 median	-0.146 MAE 1.441 strength	-0.006 MAE 0.048 bit mean	+2.477 MAE 1.294 bit mean
Prune interp d=.9-.6 Prune extrap d=.55 Quantization bits ≥4 Quantization int3			

Grouped quantization · low_order_2d (v55)

	Gain	Candidate MAE	Best simple (MAE)	Source states
bit_test	+0.206	0.281	zero (0.487)	seen development states
granularity_test	+0.680	0.508	median (1.188)	seen development states
joint_test	+0.004	0.237	median (0.241)	held-out 1B@96k

Distillation · unseen U225 pool (6 runs, Gemma-3-1B)



*1B@96k: only d=.65/.55 and int4/int3; original v46 freeze. Cell lines: gain / candidate MAE / selected simple baseline.
Pythia cells average math, code and QA. All values are nats per native token; the distillation panel is separate.